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Practice Guidelines

2018 Guidelines of the Taiwan Society of Cardiology, Taiwan Society of Emergency Medicine and Taiwan Society of Cardiovascular Interventions for the management of non ST-segment elevation acute coronary syndrome[☆]



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In Taiwan, the incidence of non-ST segment elevation acute coronary syndrome (NSTEMI-ACS) continues to increase in recent years. The purpose of this guideline is to help health care professionals in Taiwan to use adequate tests and treatments for management of NSTEMI-ACS. For rapid diagnosis, in addition to history and physical examination, 0/3 h rapid diagnosis protocol with high sensitivity cardiac troponin assay is recommended in this guideline. Dual antiplatelet and anticoagulation therapies are important parts in the initial treatment. Risk stratification should be performed to identify high risk patients for early coronary angiography. Through evaluation of the coronary anatomy and other clinical factors, the decision for coronary revascularization, either by percutaneous coronary intervention or coronary artery bypass grafting, should be decided by the heart team. The duration of dual antiplatelet therapy should be given for at least 12 months after discharge. Other secondary preventive medications are also recommended for long term use.

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Introduction

Purpose of the guideline

Acute coronary syndrome (ACS) is an important public health issue in Taiwan. Although the overall incidence of acute myocardial infarction (MI) reached a plateau phase after 2008 in Taiwan, the cases of non-ST-segment elevation MI (NSTEMI) continued to increase progressively and outnumbered ST-segment elevation MI (STEMI) from 2004.^{1,2} In all acute MI patients in 2011, 67% were NSTEMI in Taiwan.² NSTEMI-ACS includes not only NSTEMI but also unstable angina with myocardial ischemia without myocyte necrosis. The pathophysiological mechanisms of NSTEMI-ACS are complex. It usually comes from rupture or erosion of coronary atherosclerotic plaques and results in acute thrombus formation with intracoronary stenosis or obstruction. However, systemic factors, such as sepsis, anemia, respiratory failure, hypotension or arrhythmia, which cause myocardial oxygen demand and supply imbalance also may induce NSTEMI-ACS. In recent years, guidelines regarding cardiovascular (CV) diseases management have been published by the Taiwan Society of Cardiology and cause great impact on daily clinical practice in Taiwan. Similarly, the major purpose of this

guideline is to perform an extensive review of the literatures related to NSTEMI-ACS, reorganize the information and make recommendations to the health care professionals about the usefulness of diagnostic tests, procedures, or treatments for NSTEMI-ACS in Taiwan. Since the clinical situations vary among individual patients and medical resources are different in hospitals, the guideline recommendations provide general principles and the appropriate decisions still depend on clinical judgments of the in-charged physicians.

Guideline developed process

In 2016, the Executive Board of Taiwan Society of Cardiology decided to develop guidelines for management of NSTEMI-ACS. The members of this writing group were selected by the chairman of the Intervention Committee of Taiwan Society of Cardiology. Before starting to draft the guideline, three expert meetings were held in December 12, 2016 in Taipei, January 14, 2017 in Taichung and March 4, 2017 in Kaohsiung. Clinical evidences were reviewed and consensus about the diagnosis and treatment of NSTEMI-ACS were achieved during the meetings. The guideline draft was completed in October 2017. From November 2017, several symposiums were held through Taiwan to review the

recommendations suggested in the draft guideline. Modifications of the draft guideline were performed according to the opinions raised in these symposiums. The full manuscript of the guideline was completed and sent to the Taiwan Society of Cardiology, Taiwan Society of Emergency Medicine and Taiwan Society of Cardiovascular Interventions for review. After final revision, the full document was submitted for publication. In order to demonstrate the intensities of recommendations and evidences, we adopted the evidence-based classification system, including class of recommendation (COR) and level of evidence (LOE) developed by the American College of Cardiology/American Heart Association. The COR is used to indicate whether a recommendation or suggestion is useful or harmful. Class I recommendations indicate they are useful and should be used. Class III recommendations indicate they are harmful and should not be done. Class II recommendations fall in between I and III. Class IIa indicates that the evidences favor the recommendations; while class IIb indicates that the recommendations are less well established (Table 1). The LOE is used to denote the evidence intensity supporting the recommendations. LOE A indicates that multiple randomized trials results support the recommendations. LOE B indicates that only one randomized trial or large-scale observation studies support the recommendations. LOE C indicates that only small studies or expert opinions suggest the recommendations (Table 2).

Pathophysiology and epidemiology

Definition and pathophysiology

ACS is referred to a spectrum of conditions which abruptly cause an unmet need of coronary blood supply to myocardium. Based on the presentations of electrocardiography (ECG), ACS is categorized into two distinct populations: STEMI and NSTEMI-ACS. Patients with ST-segment elevation on ECG usually indicate acute complete coronary occlusion that warrants aggressive therapeutic strategies for rapid coronary reperfusion. The other patients presented without persistent ST-segment elevation on ECG are defined as NSTEMI-ACS which is further divided into NSTEMI and unstable angina according to whether the cardiac biomarkers are elevated or not.³ Ischemic ECG changes, including transient ST-segment elevation, ST-segment depression, and/or T wave inversion may be present, but are not a must for a diagnosis of NSTEMI. In this guideline, the "Third Universal Definition of Myocardial Infarction" was adopted to define acute MI.⁴ In addition to the increase of cardiac biomarkers, acute MI could be diagnosed with at least one of the following: (1) symptoms of myocardial ischemia; (2) new or presumed new significant ST-T wave changes or left bundle branch block (LBBB); (3) development of pathological Q waves; (4) imaging evidence of new or presumed new loss of viable myocardium or regional wall motion abnormality; (5) intracoronary thrombus detected on angiography or autopsy. The causes of myocardial damage are multifactorial and 5 types of acute MI are defined. They are type 1: spontaneous MI, type 2: MI secondary to ischemic imbalance, type 3: MI resulting in death when biomarker is unavailable, type 4: MI related to percutaneous coronary

intervention (PCI) or stent thrombosis and type 5: MI related to coronary artery bypass graft (CABG) surgery (Table 3). Most of the suggestions for treatment in this guideline can be applied to type 1 MI. For other types of MI, adequate management should depend on the physicians' clinical judgment due to the complicated clinical conditions in these patients.

The fundamental cause of ACS is the deficiency of myocardial oxygen supply relative to oxygen demand which is usually related to coronary artery stenosis or obstruction leading to myocardial ischemia and/or necrosis. Although numerous etiologies could bring myocardial damage, the majority of the NSTEMI-ACS begins with the process of atherosclerosis. Even though the progress is slow, it may be facilitated by plaque disruption with formation of a non-occlusive intraluminal thrombus or plaque hemorrhage. Symptomatic coronary occlusion is mainly caused by intraluminal thrombus formation induced by plaque rupture (type 1 MI).⁵ NSTEMI-ACS usually has a severe blockage rather than total occlusion in one or more coronary arteries and most often caused by acute intraluminal coronary thrombus formation.

Recommendation

- **Diagnosis of NSTEMI-ACS is mainly based on clinical symptoms, ECG and cardiac biomarker. (COR I, LOE C)**

Epidemiology of NSTEMI-ACS in Taiwan

The epidemiological study using the Taiwan National Health Insurance Research Database (NHIRD) showed that the incidence of NSTEMI increased progressively in Taiwan. Overall, 144,634 patients were hospitalized for acute MI during the 15 years between 1997 and 2011.² The number of NSTEMI case increased from 2150 cases per year in 1997 to 8603 cases per year in 2011 with a relative increase of approximately 300%.² The incidence of NSTEMI increased from 11 cases per 100,000 person-years to 28 cases per 100,000 person-years. The proportion of NSTEMI patients among all patients admitted for acute MI increased from 36.4% to 66.8% in this time period (Fig. 1).² In patient characteristics, NSTEMI patients were older (68 ± 14 years) and the male to female ratio in NSTEMI patients was approximately 2:1. NSTEMI patients also had a greater number of comorbidities, including hypertension, diabetes mellitus (DM), cerebral vascular occlusive disease, peripheral artery disease (PAD), chronic kidney disease (CKD), and heart failure, compared to STEMI patients.^{2,6} Considering treatment modalities in Taiwan, the number of NSTEMI patients that underwent revascularization therapy, including PCI and CABG, increased dramatically from 10% (9% PCI; 1% CABG) in 1997 to 56% (52% PCI; 4% CABG) in 2011.² The in-hospital mortality rate of NSTEMI decreased from 12.2% in 1997 to 7.2% in 2011. In general, NSTEMI patients tended to be hospitalized for a longer duration (10 ± 15 days) than that in STEMI patients (8 ± 12 days).^{2,6} The Taiwan ACS Full Spectrum Registry study recruited 3183 ACS patients from 39 hospitals around Taiwan between October 2008 and January 2010. Among the included patients, 34% of them had NSTEMI-ACS.⁷ Compared to STEMI patients, those with NSTEMI-ACS presented lower rates of in-

Table 1 Class of recommendation.

Class of recommendation	Definition	Strength
Class I	Evidence or general agreement that a given treatment is beneficial, useful, effective.	Is recommended/is indicated
Class II	Conflicting evidence or a divergence of opinion about the usefulness or efficacy of a treatment.	
Class IIa	Weight of evidence/opinion is in favor of usefulness or efficacy	Should be considered
Class IIb	Usefulness or efficacy is less well established by evidence/opinion.	May be considered
Class III	Evidence or general agreement that the given treatment is not useful or effective and in some cases may be harmful.	Is not recommended

hospital death, new onset of cardiac shock, ventricular arrhythmias and atrial fibrillation (AF) during hospitalization.^{7,8} However, NSTEMI-ACS patients had a significantly higher risk of mortality (10.1% vs 6.1%) and reattack of MI (6.4% vs 4.7%) than STEMI patients at 1-year follow up after discharge. NSTEMI patients were also more likely to receive CABG (2.6% vs 1.0%).⁸

Recommendation

- It should be noted that NSTEMI is increasing in Taiwan and carries a higher morbidity and mortality than STEMI after discharge. (COR I, LOE B)

Diagnosis

History and physical examination

Clinical symptoms are important to the diagnosis of NSTEMI-ACS. During history taking, patients with NSTEMI-ACS frequently complain of chest pain and commonly use the terms such as pressure, tightness or heaviness to describe their chest symptoms. The chest pain is usually retrosternal in location, spreading frequently to both sides of the anterior chest (especially left chest) with radiation to left shoulder, arm, neck, and jaw. Less commonly, the pain is epigastric, leading the patient and physician to mistake it for indigestion. Rarely, ischemic chest pain may be noted on the right side of the chest or the interscapular region. The duration of

chest pain usually lasts for more than 10 min. Severe pain that radiates through the chest into the back is highly suggestive of aortic dissection. Pain that the patient can localize by pointing with one finger or brief episodes of pain that last a few seconds or less is rarely ischemic. The associated symptoms of NSTEMI-ACS include dyspnea, cold sweating, nausea and vomiting. Other atypical symptoms include anxiety, feelings of profound weakness, dizziness, syncope, stabbing or pleuritic pain and palpitation. Atypical symptoms are more common among elderly patients, women and in patients with DM, CKD or dementia.^{9,10}

The physical findings in NSTEMI-ACS vary from completely unremarkable to dramatic, depending on the degree and location of myocardial ischemia or necrosis. A third heart sound can be heard in patients with left ventricular systolic dysfunction; the fourth heart sound occurs in patients with acute myocardial ischemia. Rarely, a systolic murmur may result from mechanical complications (papillary muscle dysfunction or rupture or ventricular septal defect). The findings of physical examination are helpful and important for differential diagnosis in patients presenting with acute chest pain. A pulsatile abdominal mass suggests abdominal aortic aneurysm. Aortic dissection should be suspected when there are blood pressure or pulse disparities (unequal palpated pulse volume or a difference of systolic blood pressure ≥ 15 mmHg between both arms) or a new murmur of aortic regurgitation accompanied by back pain. Pericardial friction rubs may be heard in patients with pericarditis or pericardial effusion. Differences in breathing sounds in the presence of acute dyspnea and pleuritic chest pain raise the possibility of pneumothorax. A more than 10 mmHg fall in systolic pressure with inspiration (pulsus paradoxus) is considered a sign of pericardial or pulmonary disease. It is not specific for pericardial tamponade and can accompany massive pulmonary embolism or tension pneumothorax.

Electrocardiogram

The resting 12-lead ECG is the most important initial diagnostic test in the assessment of patients with suspected NSTEMI-ACS and should be performed as soon as possible, preferably within 10 min of initial medical contact and immediately interpreted by a qualified physician.¹¹

Table 2 Level of evidence.

Level of evidence A	Data derived from multiple randomized clinical trials
Level of evidence B	Data derived from a single randomized clinical trial, meta-analyses, or large non-randomized studies
Level of evidence C	Subgroup analyses, post-hoc analyses, retrospective studies, cohort studies, registries, small studies, or expert opinion

Table 3 Clinical classification of myocardial infarction.**Type 1: Spontaneous MI**

Spontaneous MI related to atherosclerotic plaque rupture, fissuring, or erosion with resulting coronary intraluminal thrombus formation.

Type 2: MI secondary to an ischemic imbalance

Myocardial injury with conditions of imbalance between myocardial oxygen supply and demand. For example, coronary artery spasm or embolism, arrhythmias, anemia, respiratory failure, hypotension and hypertension.

Type 3: MI resulting in death when biomarker values are unavailable

Cardiac death with symptoms suggestive of myocardial ischemia but death occurring before blood samples could be obtained.

Type 4a: MI related to PCI

MI associated with PCI is arbitrarily defined by elevation of cTn values $>5\times$ 99th percentile URL in patients with normal baseline values or a rise of cTn values $>20\%$ if the baseline values are elevated.

Type 4b: MI related to stent thrombosis

MI associated with stent thrombosis is detected by coronary angiography or autopsy with a rise and/or fall of cardiac biomarkers values with at least one value above the 99th percentile URL.

Type 5: MI related to CABG

MI associated with CABG is arbitrarily defined by elevation of cardiac biomarker values $>10\times$ 99th percentile URL in patients with normal baseline cTn values. In addition, either (i) new pathological Q waves or new LBBB, or (ii) angiographic documented new graft or new native coronary artery occlusion, or (iii) imaging evidence of new loss of viable myocardium or new regional wall motion abnormality.

CABG, coronary artery bypass grafting; cTn, cardiac troponin; LBBB, left bundle branch block.

PCI, percutaneous coronary intervention; URL, upper reference limit.

Adapted from reference.⁴

Characteristic ECG change in patients with NSTEMI-ACS include ST depression, transient ST elevation, new T wave inversion or normalization of a previously negative T wave.¹² New (or presumably new) ST-segment deviation (≥ 0.1 mV) is a useful measure of ischemia and prognosis. ST segment depression in anterior precordial leads (V1-4) with R/S wave ratio greater than 1 in leads V1 or V2 indicates

true posterior MI and should be treated according to the STEMI. The location of ST-segment depression on the ECG is less specific for determining the culprit coronary artery. The presence of a normal initial ECG does not completely exclude the diagnosis of NSTEMI-ACS. The ECG should be repeated at 15- to 30-min intervals during the first hour, especially for those patients with signs or symptoms

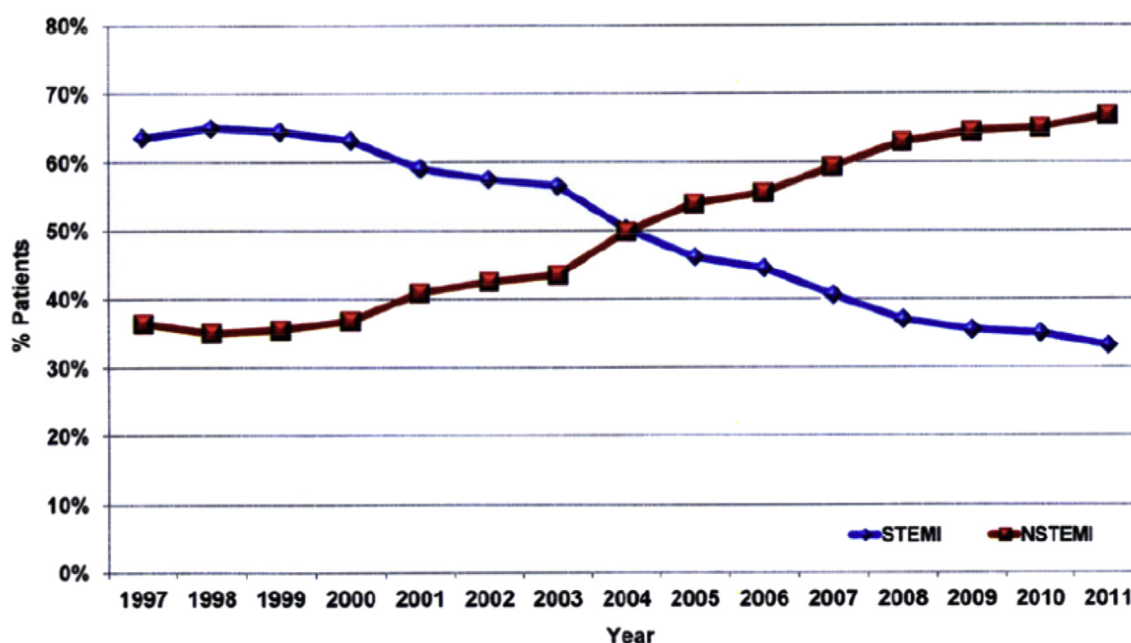


Figure 1 The percentage of patients admitted for STEMI and NSTEMI by study years in Taiwan. NSTEMI, non-ST-segment elevation myocardial infarction; STEMI, ST-segment elevation myocardial infarction. Adapted from reference.²

suggestive of ongoing myocardial ischemia. Additional leads could be recorded because some left circumflex artery occlusion is detected only in V7–V9 and right ventricular MI is detected in the right-sided leads (V3R to V4R).

Recommendation

- In patients with suspected NSTEMI-ACS, ECG should be performed within 10 min of initial medical contact and interpreted by a qualified physician. (COR I, LOE C)
- ECG should be repeated at 15- to 30-min intervals during the first hour for those patients with signs or symptoms suggestive of ongoing myocardial ischemia. (COR I, LOE C)

Cardiac biomarker

The most sensitive and specific biomarkers for detecting myocardial necrosis are cardiac troponin (cTn) T or cTn I. Currently, high sensitivity cardiac troponin (hs-cTn) assays have replaced traditional assays gradually and confer a $\geq 99\%$ negative predictive value for diagnosing acute MI.^{13–15} The criteria to define hs-cTn assays include: (1) the detectable concentrations below the 99th percentile value are at least 50% (and ideally $> 95\%$) of healthy individuals; and (2) the coefficient of variation at the 99th percentile value is $< 10\%$. As the sensitivity of assays increases, elevated cTn could be found in several conditions other than acute MI, such as heart failure, CKD, arrhythmia or pulmonary embolism (Table 4).¹⁶ Thus, a single elevation of hs-cTn is usually not enough and serial changes of hs-cTn for the diagnosis of acute MI become necessary and provide more information. In patients with symptoms of acute myocardial ischemia, a highly elevated hs-cTn above the 99th percentile of healthy individuals could suggest acute MI if the baseline value is normal. A serial increase or

decrease $\geq 20\%$ is rather crucial to confirm the diagnosis of acute MI if the initial value is elevated.¹⁷ Usually, troponins rise as early as 2–4 h (within 1 h when using hs-cTn assays) after symptom onset in acute MI.^{14,17} In majority of patients with symptoms or ECG changes suggestive of ACS, MI can be excluded or identified within 6 h after onset of symptoms by using cTn assays.

Several multicenter studies have shown that hs-cTn assays not only increase diagnostic accuracy for acute MI, but also allow for a rapid rule out of acute MI in patients presenting early after chest pain onset.^{17,18} After considering local circumstances in Taiwan, a 0 h/3 h rapid diagnosis protocol with hs-cTn assay is recommended in this guideline (Fig. 2). In patients presenting with acute chest pain and suspected ACS, acute MI should be ruled in the following 3 conditions: (1) if the initial hs-cTn levels are $< 99\text{th}$ percentile of healthy controls and become above the 99th percentile and the delta change (value at 3 h – value at 0 h/value at 0 h) $> 50\%$ after 3 h; (2) if the initial hs-cTn levels are $> 99\text{th}$ percentile and with a delta change (value at 3 h – value at 0 h/value at 0 h) $> 20\%$ after 3 h; (3) if the patients have typical ischemic symptoms, ECG changes and the initial hs-cTn levels are highly abnormal ($> 5\text{-fold}$ the 99th percentile). These patients should be recognized as high risk group and further management becomes necessary. Acute MI could be ruled out if the hs-cTn levels at initial and 3 h later are all $< 99\text{th}$ percentile of healthy controls. For other conditions that do not fulfill rule in or rule out criteria, the patients may need observation and further examinations with possible differential diagnosis should be made. Increased hs-cTn could occur in other cardiac or non-cardiac conditions (Table 4). CKD is a common dilemma to use cTn because CKD usually have chronic elevated cTn which may reduce its specificity and require extended evaluation for diagnosing acute MI. The National Academy of Clinical Biochemistry suggests a rise and fall in cTn levels greater than 20% with at least 1 value exceeding the 99th percentile to make an acute MI diagnosis among patients with end-stage renal disease.¹⁹ Overall, the 0 h/3 h rapid diagnosis protocol should be used in conjunction with detailed assessment of symptoms, ECG or other necessary tests to avoid over- or under-diagnosis of acute MI and miss other acute life-threatening diseases causing chest pain. Creatine kinase isoenzyme MB (CK-MB) has been the most commonly used biomarkers to diagnose acute MI for decades. When compared to cTn, however, CK-MB is much less sensitive for detecting myocardial injury because more tissue injury is required for the elevation of CK-MB. Beyond serving as a biomarker for diagnosing MI, CK-MB could provide information to estimate the size of MI. It also indicates the timing of myocardial injury due to its rapid rise and decline after MI when compared with cTn. Unstable angina is diagnosed when there is no evidence of cardiomyocyte necrosis.

Recommendation

- Hs-cTn assays are recommended in all patients suspected of ACS. (COR I, LOE B)
- By using hs-cTn assay, the 0 h/3 h rapid diagnosis protocol is recommended. (COR I, LOE B)
- CK-MB is a complementary examination to cTn if clinical condition is necessary for diagnosis of ACS. (COR I, LOE B)

Table 4 Conditions associated with cardiac troponin elevation other than acute myocardial infarction.

Tachyarrhythmias
Heart failure
Hypertensive emergencies
Shock, sepsis or burn
Myocarditis
Takotsubo cardiomyopathy
Aortic dissection
Pulmonary embolism, pulmonary hypertension
Renal dysfunction
Coronary spasm
Acute stroke or subarachnoid hemorrhage
Cardiac contusion or cardiac procedures
Hypothyroidism and hyperthyroidism
Myocardial infiltrative diseases
Myocardial drug toxicity or poisoning (e.g., doxorubicin, 5-fluorouracil, herceptin, snake venoms)
Extreme endurance efforts
Rhabdomyolysis

Adapted from reference.¹⁶

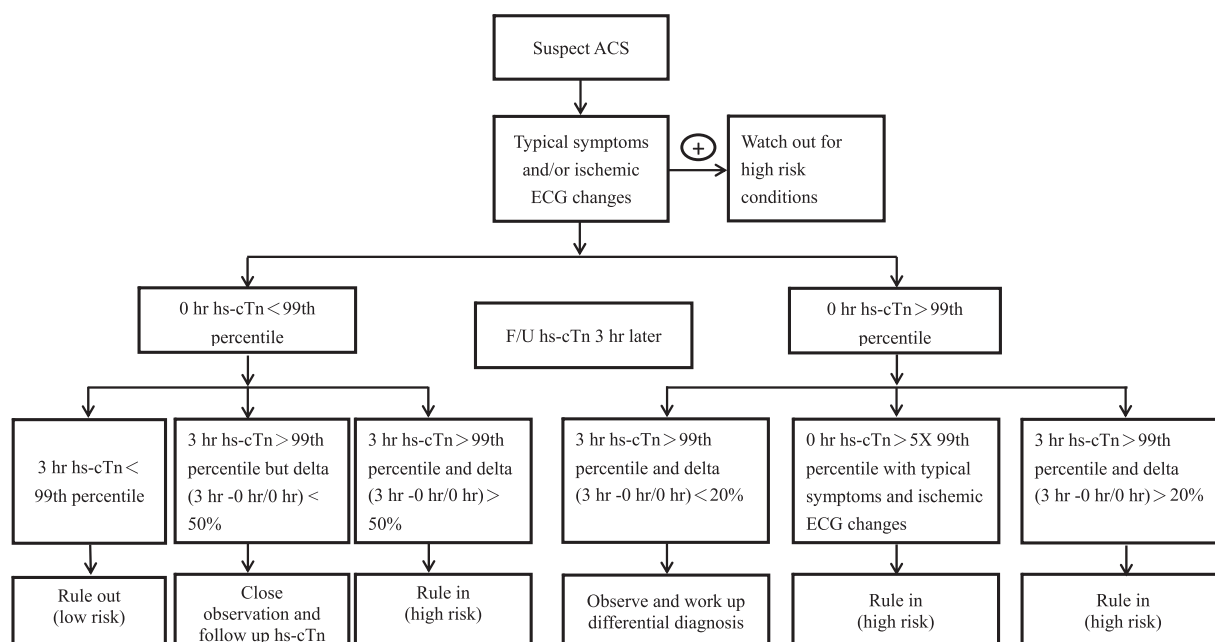


Figure 2 The 0 h/3 h rapid diagnosis protocol with hs-cTn assay. ACS, acute coronary syndrome; ECG, electrocardiogram; hs-cTn, high sensitivity cardiac troponin.

Non-invasive cardiac imaging

Transthoracic echocardiography is the most commonly used imaging modality and should be routinely performed in all patients during hospitalization for NSTEMI-ACS. In addition to evaluate left ventricular function and identify regional wall motion abnormality, echocardiography is helpful in detecting other cardiac or non-cardiac diseases that may cause chest pain, such as acute aortic dissection, pericardial effusion, valvular heart disease, hypertrophic cardiomyopathy or right ventricular dysfunction caused by pulmonary embolism. Coronary computed tomography angiography (CTA) is another imaging modality that can be considered in those patients with inconclusive ECG and cTn results. CTA is a safe and rapid imaging technique to exclude the presence of severe coronary stenosis. A meta-analysis including 4 randomized controlled trials showed that coronary CTA decreased the cost of emergency department and length of hospital stay in patients presenting with acute chest pain. However, there was no significant difference of the rate of death and MI between the patients evaluated with and without CTA.²⁰ In addition to CTA, stress test is another reasonable choice for those patients with low risk combined with negative cardiac biomarkers. In such kind of patients, immediate exercise testing is safe and has high negative predictive value for future cardiac event.²¹ Stress echocardiography or myocardial perfusion imaging also showed prognostic value in low risk patients with negative cTn.^{22,23}

Recommendation

- Transthoracic echocardiography should be performed in patients with NSTEMI-ACS. (Class I, LOE C)
- Coronary CTA could be considered to exclude severe coronary stenosis in patients with inconclusive ECG and cTn results. (Class IIa, LOE B)

- Stress tests (exercise testing, stress echocardiography, stress myocardial perfusion imaging) could be considered and have prognostic value in those patients with low risk combined with negative cardiac biomarkers. (Class IIa, LOE B)

Risk assessment

Risk assessment or stratification is usually the first step after the diagnosis of NSTEMI-ACS. Different risk category carries different risk of death and complications in its natural clinical course. Traditionally, physicians make risk assessment from history, physical examination, ECG, chest film and laboratory findings. Nowadays, there are established ischemic risk scores (Thrombolysis in Myocardial Infarction [TIMI] risk score and Global Registry of Acute Coronary Events [GRACE] risk score) that can be used to perform global risk assessment in patients with NSTEMI-ACS.^{24,25} The TIMI risk score (range 0–7) consists of seven dichotomous variables from age to biomarker (Table 5). Low, intermediate and high risk is defined as TIMI risk score 0–2, 3–4 and 5–7. In low risk patients, the risk of death,

Table 5 The TIMI risk score.

1. Age ≥ 65 years	1 point
2. ≥ 3 risk factors for CAD	1 point
3. Use of aspirin in last 7 days	1 point
4. Known CAD (prior stenosis $\geq 50\%$)	1 point
5. >2 anginal events in last 24 h	1 point
6. ST-segment deviation	1 point
7. Elevated cardiac markers	1 point

CAD, coronary artery disease; TIMI, Thrombolysis in Myocardial Infarction.

Adapted from reference.²⁴

Table 6 The GRACE risk score.

killip class	Points	SBP, mm Hg	Points	Heart rate, Beats/min	Points	Age, yr	Points	Creatinine, mg/dL	Points
I	0	≤80	58	≤50	0	≤30	0	0–0.39	1
II	20	80–99	53	50–69	3	30–39	8	0.40–0.79	4
III	39	100–119	43	70–89	9	40–49	25	0.8–1.19	7
IV	59	120–139	34	90–109	15	50–59	41	1.20–1.59	10
		140–159	24	110–149	24	60–69	58	1.60–1.99	13
		160–199	10	150–199	38	70–79	75	2.00–3.99	21
		≥200	0	≥200	46	80–89	91	> 4.0	28
						≥90	100		
Other factors									Points
Cardiac arrest at admission									39
ST-segment deviation									28
Elevated cardiac enzyme level									14

GRACE, Global Registry of Acute Coronary Events; SBP, systolic blood pressure.

Adapted from reference.²⁵

new or recurrent MI, or severe recurrent angina requiring urgent revascularization within 14 days is less than 10%; while the risk is 13–20% in intermediate risk group and 26–41% in high risk group.²⁴ GRACE score is more complex and uses eight parameters to calculate the risk including age, heart rate, systolic blood pressure, Killip class, serum creatinine level, cardiac arrest at hospital admission, ST-segment deviation in ECG and elevated cardiac marker (Table 6).²⁵ Several studies compared the discriminative abilities between the TIMI and GRACE risk scores.^{26,27} Overall, the TIMI and GRACE scores are both useful in identifying high risk patients with NSTEMI-ACS. The GRACE score is more complex and the TIMI score is much easier to use in the acute settings.

Recommendation

- It is recommended to perform risk assessment in NSTEMI-ACS patients based on clinical history, vital signs, physical examination, ECG and laboratory findings. (COR I, LOE C)
- The established scoring systems, such as TIMI or GRACE risk scores can be used for risk assessment. (COR I, LOE B)

General treatment

Oxygen

Oxygen inhalation has been shown to reduce infarction size in experimental coronary artery occlusion study.²⁸ However, randomized controlled trials for STEMI patients without hypoxia showed oxygen supplement was not associated with clinical benefits and may even increase early infarction injury and lead to larger infarction size.^{29–31} The efficacy and safety of supplemental oxygen in NSTEMI-ACS patients remain unclear. The Determination of the Role of Oxygen in Suspected Acute Myocardial Infarction (DETO2X-AMI) study included both STEMI and NSTEMI patients and found that routine use of supplemental oxygen in patients with suspected MI had no benefit in reducing 1-month and

1-year all-cause mortality.³² Routine oxygen supplement to NSTEMI-ACS patients without hypoxia is not recommended. Supplemental oxygen is indicated if blood oxygen saturation is less than 90% or if there is evidence of respiratory distress.

Recommendation

- Supplemental oxygen is indicated if blood oxygen saturation is less than 90% or if there is evidence of respiratory distress. (COR I, LOE A)

Nitrate

Nitrates cause peripheral and coronary dilatation in an endothelium-independent way which improves coronary blood flow, reduces preload/afterload of left ventricle and relieves ischemic symptoms.^{33–35} The indications of nitrates are mainly for symptomatic relief of angina or heart failure and control of hypertension. Routine prescription of nitrates for asymptomatic patients is not recommended.³⁶ For patients with persistent ischemic symptoms, sublingual NTG (0.4 mg) may be used every 5 min for maximal 3 times if necessary. In patients with ongoing angina, intravenous nitrates are more effective to relieve symptoms.^{37,38} The dose should be up-titrated periodically until the therapeutic effect is achieved or when systolic blood pressure is less than 90 mmHg. Common adverse effects of nitrates include dizziness, throbbing headache, or hypotension. Nitrates should be avoided in patients with systolic blood pressure <90 mmHg and are contraindicated in patients who recently received a phosphodiesterase type 5 inhibitor (within 24 h for sildenafil or vardenafil and 48 h for tadalafil).³⁹

Recommendation

- Repeated sublingual nitroglycerin every 5 min for maximal 3 doses is recommended in NSTEMI-ACS patients as initial therapy to relieve ischemic symptoms. (COR I, LOE C)

- Intravenous nitroglycerin is indicated for NSTEMI-ACS patients with refractory angina, uncontrolled hypertension or symptoms of heart failure. (COR I, LOE B)
- Nitrate is not recommended in patients who recently received a phosphodiesterase type 5 inhibitor. (COR III, LOE B)

Beta-blocker

Beta-blockers reduce myocardial oxygen demand by decreasing myocardial contractility, blood pressure, and heart rate. They are shown to be associated with decreased myocardial ischemia, re-infarction, and the risk of ventricular fibrillation in acute MI patients.^{40–42} Meta-analysis also demonstrated that beta-blockers reduce in-hospital mortality and the risk of other adverse CV outcomes.^{43,44} Oral beta-blockers were also shown to improve left ventricular function and in-hospital survival rate if they are administered early within 24 h after admission for ACS.⁴⁵ However, early use of beta-blockers, especially the intravenous form, may be associated with increased risk of cardiogenic shock in both STEMI and NSTEMI patients with age greater than 70 years, systolic blood pressure less than 120 mmHg, or sinus tachycardia greater than 110 bpm.^{46,47} Beta-blockers should also be avoided in patients with other contraindications such as PR interval greater than 0.24 s, second- or third-degree heart block, evidence of low output state, active asthma, or reactive airway disease. However, beta-blockers are not contraindicated for patients with chronic obstructive pulmonary disease if active bronchospasm is not present. Due to the benefits of mortality reduction in patients with systolic heart failure, beta-blockers without intrinsic sympathomimetic activity such as bisoprolol, carvedilol, or extended-released metoprolol succinate should be prescribed before discharge in patients with impaired left ventricular systolic function.^{48–50} For NSTEMI-ACS patients with coronary spasm or who take cocaine, beta-blockers should be avoided to prevent the unopposed alpha-mediated vasoconstriction.

Recommendation

- Early administration of beta-blockers within 24 h of hospitalization is recommended for NSTEMI-ACS patients without contraindications and who have no risk factors for shock (COR I, LOE A).
- Early administration of intravenous beta-blockers may be harmful and is not recommended for patients with risk factors of cardiogenic shock (COR III, LOE B).

Calcium channel blocker

Calcium channel blockers (CCBs), including dihydropyridine and non-dihydropyridine, lead to coronary vasodilatation and improve coronary blood flow. Therefore, they are preferred in NSTEMI-ACS patients with coronary spasm.⁵¹ The non-dihydropyridines (diltiazem and verapamil) suppress myocardial contractility, slow down heart rate and delay AV nodal conduction, all of which could improve angina symptoms.^{52,53} In acute MI patients, some but not all studies have shown the non-dihydropyridines have beneficial effects on re-infarction size reduction and long-term outcomes improvements.^{54–57} Non-dihydropyridines calcium channel blockers can be prescribed for NSTEMI-ACS

patients with ischemic symptoms and are contraindicated for beta-blockers, if there is no contraindication such as significant left ventricular dysfunction, cardiogenic shock, or high degree atrioventricular block. However, short-acting nifedipine has been demonstrated to increase mortality rate in patients with coronary artery disease (CAD).⁵⁸ Another study of unstable angina patients demonstrated that the combination of immediate-release nifedipine to metoprolol provide no further benefit, and nifedipine alone was associated with a trend to increase the MI risk.⁵⁹ Therefore, short acting nifedipine is not recommended in NSTEMI-ACS patients. Data from large clinical trials about the efficacy of other dihydropyridines in NSTEMI-ACS patients are lacking.

Recommendation

- Oral non-dihydropyridine CCBs are recommended for NSTEMI-ACS patients with ischemic symptoms who are contraindicated for beta-blockers (COR I, LOE B).
- It is reasonable to give oral non-dihydropyridine CCBs to NSTEMI-ACS patients who have refractory angina under adequate use of beta-blockers and nitrates (COR IIa, LOE C).
- CCBs and nitrates instead of beta-blockers should be considered for NSTEMI-ACS patients with coronary vasospasm (COR IIa, LOE B).
- Short-acting nifedipine is not recommended in NSTEMI-ACS patients (COR III, LOE B).

Renin-angiotensin-aldosterone system inhibitor

In patients with acute MI, early administration of angiotensin converting enzyme inhibitors (ACEIs) have beneficial effects on total mortality reduction.^{60,61} The survival benefit of ACEIs was consistent in MI patients with left ventricular dysfunction or heart failure.^{62–64} In NSTEMI patients, post-hoc analysis of the Survival of Myocardial Infarction Long-term Evaluation (SMILE) study demonstrated zofenopril treatment significantly reduced one year mortality rate.⁶⁵ A meta-analysis also proved the survival improvement in acute MI patients with early use of ACEI within 36 h of disease attack.⁶⁶ Angiotensin receptor blocker (ARB) has been shown to have similar CV protection effects as ACEI for MI patients in the Valsartan in Acute Myocardial Infarction (VALIANT) trial.⁶⁷ However, the combination of ACEI and ARB not only failed to show survival benefit but increased the risk of adverse events in this study.⁶⁷ In acute MI patients with left ventricular dysfunction and heart failure, the addition of eplerenone, a selective aldosterone blocker, was shown to be effective in reducing morbidity and mortality.⁶⁸

Recommendation

- ACEIs are recommended for all NSTEMI-ACS patients during hospitalization unless contraindicated (COR I, LOE A).
- ARBs are recommended for all NSTEMI-ACS patients who are ACEIs intolerant during hospitalization unless contraindicated (COR I, LOE A).
- Aldosterone blockade, preferably eplerenone, is recommended for NSTEMI-ACS patients without significant renal dysfunction or hyperkalemia, who have received

optimal therapy with ACEI and beta-blocker and have left ventricular ejection fraction < 40% with heart failure (COR I, LOE A).

- The combination of ACEI and ARB is not recommended in NSTEMI-ACS patients (COR III, LOE A)

Lipid lowering therapy

In ACS patients, lipid lowering therapy with statin or statin plus ezetimibe to lower low density lipoprotein cholesterol (LDL-C) has been proved to improve the clinical outcomes. The Myocardial Ischemia Reduction with Aggressive Cholesterol Lowering (MIRACL) study randomized ACS patients into atorvastatin 80 mg daily and placebo between 24 and 96 h after hospitalization. After 16 weeks of follow-up, 16% reduction of ischemic events (95% confidence interval [CI], 0.70–1.00; $p = 0.048$) was observed in statin group.⁶⁹ In the Pravastatin or Atorvastatin Evaluation and Infection Therapy–Thrombolysis in Myocardial Infarction 22 (PROVE IT–TIMI 22) study, among patients with ACS, atorvastatin 80 mg daily provided 16% risk reduction in primary endpoint (95% CI 0.05–0.26, $p = 0.005$) compared with pravastatin 40 mg daily after an average follow-up of 2 years.⁷⁰ In the Improved Reduction of Outcomes: Vytorin Efficacy International Trial (IMPROVE-IT) study, compared with simvastatin monotherapy, adding ezetimibe to simvastatin 40 mg daily in ACS patients resulted in further LDL-C reduction from 70 to 53 mg/dL.⁷¹ After 7 years of follow-up, the combination of ezetimibe and simvastatin provided additional 6.4% primary endpoint risk reduction (95% CI 0.89–0.99, $p = 0.016$) comparing with simvastatin alone and the benefits were more prominent in diabetic patients. A nationwide population-based cohort study using Taiwan NHIRD showed that, during 6-year follow-up, ACS patients receiving statin and ezetimibe combination therapy had lower risk of re-hospitalization and revascularization than those receiving statin alone.⁷² In Japan, in-hospital initiation of statin therapy immediately after ACS attack with atorvastatin 20 mg daily not only reduced coronary plaque volume 6 months later, but also resulted in long-term CV risk reduction after 4 years follow-up.^{73,74} Based on these evidences, the Taiwan lipid guideline for high risk patients suggests that the LDL-C target is < 70 mg/dL for NSTEMI-ACS patients and a lower target of 55 mg/dL can be considered for NSTEMI-ACS patients with DM.⁷⁵

Recommendation

- Statin or statin/ezetimibe combination is recommended for all NSTEMI-ACS patients during hospitalization unless contraindicated (COR I, LOE A).
- The LDL-C target is < 70 mg/dL for NSTEMI-ACS patients. (COR I, LOE B).
- A lower LDL-C target of < 55 mg/dL can be considered for NSTEMI-ACS patients with diabetes. (COR IIa, LOE B).

Antiplatelet therapy

Aspirin

In randomized control trials^{76–79} and meta-analysis,^{80,81} aspirin was shown to have consistent effects on reducing

the incidence of MI or death in patients with NSTEMI-ACS. Aspirin should be prescribed promptly in the emergency department for all ACS patients without taking aspirin before if there are no contraindications. The contraindications of aspirin include active gastrointestinal bleeding, known hypersensitivity, severe hepatic disease, and known coagulopathy. The recommended loading dose of aspirin in Taiwan is 300 mg and the non-enteric-coated chewable form is preferred because it is absorbed more rapidly. A maintenance daily dose of 100 mg should be continued indefinitely. Aspirin maintenance dose higher than 100 mg/day does not gain more benefit⁸² and the gastrointestinal side effects increase significantly if the dose is higher than 160 mg/day.⁸³ The therapeutic benefits were uncertain if the maintenance dose is lower than 75 mg/day.⁸¹ Non-steroidal anti-inflammatory drugs should be avoided in patients who take aspirin because they reversibly bind to cyclooxygenase-1 receptor and prevent the antiplatelet effect provided by aspirin, which may cause pro-thrombosis.

Recommendation

- For patients with NSTEMI-ACS, aspirin is recommended at an initial oral loading dose of 300 mg (in aspirin-naïve patients) and a maintenance dose of 100 mg/day if there is no contraindication. (COR I, LOE A)

P2Y12 inhibitor

Clopidogrel

Concomitant use of aspirin and clopidogrel in NSTEMI-ACS patients has been shown to reduce major CV events compared with aspirin alone in the Clopidogrel in Unstable Angina to Prevent Recurrent Events (CURE) study.⁷⁹ The beneficial effect of dual antiplatelet therapy (DAPT) to reduce the ischemic events was consistent in NSTEMI-ACS patients undergoing PCI.⁸⁴ In Taiwan, 2 observational studies of ACS patients showed DAPT with aspirin and clopidogrel for more than 9 months was associated with reduced CV risk.^{8,85} Higher clopidogrel loading with 600 mg provided more rapid and better platelet inhibition compared with 300 mg loading in patients receiving elective PCI.^{86,87} In the Clopidogrel and Aspirin Optimal Dose Usage to Reduce Recurrent Events—Seventh Organization to Assess Strategies in Ischemic Symptoms (CURRENT–OASIS 7) trial, clopidogrel 600 mg loading dose was associated with risk reduction of CV events and stent thrombosis in patients who underwent PCI.⁸⁸ A meta-analysis comparing 600 mg and 300 mg clopidogrel loading regimens in patients undergoing PCI showed intensified clopidogrel loading reduced the risk of adverse CV events without increasing major bleeding.⁸⁹ However, in the real world, it is reasonable to select a lower loading dose for patients who have clinical features of high bleeding risk. Based on these considerations, a loading dose of 300 mg–600 mg is recommended in patients with NSTEMI-ACS in Taiwan. Clopidogrel is an inactive prodrug which requires 2 steps of metabolism to become active metabolite.⁹⁰ Genetic polymorphisms may contribute to the clopidogrel response variability which is associated with clopidogrel resistance in acute MI patients and increases the risk of CV events

independently.^{91,92} Ticlopidine is the first generation thienopyridine. Due to its notorious side effect of neutropenia and thrombocytopenia, it is currently used only if patients are allergic to clopidogrel and other newer P2Y12 inhibitors.

Prasugrel

Prasugrel (60 mg loading and 10 mg/day maintenance dose) is a new generation P2Y12 inhibitor that achieves greater and faster platelet inhibition comparing with clopidogrel in patients undergoing PCI.⁹³ The Trial to Assess Improvement in Therapeutic Outcomes by Optimizing Platelet Inhibition with Prasugrel—Thrombolysis in Myocardial Infarction (TRITON—TIMI) 38 trial compared prasugrel with clopidogrel in 13,608 ACS patients proceed with PCI. In the NSTEMI-ACS subgroup, prasugrel was associated with reduced ischemic events (hazard ratio [HR] 0.82, 95% CI 0.73–0.93, $p = 0.002$) compared with clopidogrel, but also increased non-CABG TIMI major bleeding (HR 1.40, 95% CI 1.05–1.88, $p = 0.02$).⁹⁴ In the TRITON-TIMI 38 overall trial, prasugrel was also shown to have net harm in patients with a history of ischemic stroke or transient ischemic attack (TIA) due to higher bleeding risk in these patients. In addition, no apparent benefit of prasugrel was shown in patient >75 years of age or with low body weight (<60 kg).⁹⁵ Therefore, prasugrel should be avoided in patients with prior history of stroke/TIA and should be used with caution in patients with old age or low body weight. In the Targeted Platelet Inhibition to Clarify the Optimal Strategy to Medically Manage Acute Coronary Syndromes (TRILOGY ACS) study, prasugrel was associated with similar ischemic and bleeding events comparing with clopidogrel among NSTEMI-ACS patients only receiving medical control.⁹⁶ The Comparison of Prasugrel at the Time of Percutaneous Coronary Intervention or as Pre-treatment at the Time of Diagnosis in Patients with Non-ST Elevation Myocardial Infarction (ACCOAST) study evaluated the pretreatment strategy of prasugrel in NSTEMI-ACS patients. It demonstrated that the pretreatment of prasugrel before catheterization did not reduce the rate of major ischemic events but increased the rate of major bleeding complications.⁹⁷ Therefore, prasugrel is not recommended for pretreatment therapy in patients with NSTEMI-ACS and should only be prescribed when PCI is planned after coronary angiography. In Japan, a lower loading and maintenance doses (20/3.75 mg) of prasugrel are used due to the concern of increased age and lower body weight of the patients. The PRASugrel compared with clopidogrel For Japanese patients with ACS undergoing PCI (PRASFIT-ACS) study showed that prasugrel was associated with a nonsignificant 23% reduction (HR 0.77, 95% CI 0.56–1.07) of adverse CV events compared with clopidogrel and the incidence of non-CABG major bleeding was similar to clopidogrel in Japanese ACS patients undergoing PCI.⁹⁸

Ticagrelor

Ticagrelor (180 mg loading and 90 mg twice daily maintenance dose) is another new generation P2Y12 inhibitor. It is an active drug and does not require metabolic transformation. Compared with clopidogrel, platelet inhibition was achieved more rapidly and greatly by ticagrelor. Due to its reversible P2Y12 binding activity, the platelet inhibition offset was faster after drug discontinuation.⁹⁹ The Platelet

Inhibition and Patient Outcomes (PLATO) trial compared ticagrelor with clopidogrel in 18,624 ACS patients, with median drug exposure duration of 9 months and treatment time up to 12 months. Compared with clopidogrel, ticagrelor significantly reduced primary composite efficacy endpoints (death from vascular causes, MI, or stroke) (HR 0.84, 95% CI 0.77–0.92, $p < 0.001$). However, ticagrelor was also associated with higher rate of major bleeding not related to CABG (4.5% vs. 3.8%, $p = 0.03$).¹⁰⁰ The benefits of ticagrelor over clopidogrel were consistent with those from the overall PLATO results both in patients initially intended for invasive or non-invasive management.^{101,102} In the NSTEMI-ACS subgroup of the PLATO trial ($n = 11,080$), the risks of ischemic events (HR 0.83, 95% CI 0.74–0.93), MI (HR 0.86, 95% CI 0.74–0.99) and CV death (HR 0.77, 95% CI 0.64–0.93) were all significantly reduced with ticagrelor. But the non-CABG major bleeding risk still increased (HR 1.28, 95% CI 1.05–1.56).¹⁰³ Data from observational studies and small randomized controlled trials comparing the efficacy and safety of ticagrelor to clopidogrel in Asian ACS patients showed controversial results.^{104–110} With similar study design to the PLATO study, the PHILO study enrolled 801 ACS patients from Japan, Taiwan, and South Korea and showed non-significantly increased risks of ischemic and bleeding risks in patients receiving ticagrelor compared with clopidogrel.¹⁰⁴ In Taiwan, two observational studies comparing ticagrelor and clopidogrel in ACS patients showed ischemic risk reduction and similar major bleeding rate in patients treated with ticagrelor.^{109,110} Based on current available evidences, ticagrelor should be the preferred P2Y12 inhibitor in NSTEMI-ACS patients treated with invasive or ischemia-guided strategy in Taiwan. However, it is reasonable to choose clopidogrel rather than newer P2Y12 inhibitors in patients with high bleeding risk features, such as old age, low body weight, anemia, CKD, concomitant use of oral anticoagulant (OAC), prior intracranial hemorrhage (ICH) or major bleeding history. The choice of P2Y12 inhibitors and their doses might be individualized to balance the ischemic and bleeding risk in NSTEMI-ACS patients in Taiwan.

Recommendation

- Ticagrelor (180 mg loading dose then 90 mg twice daily) or clopidogrel (300–600 mg loading dose, 75 mg daily dose) is recommended in NSTEMI-ACS patients treated with either invasive or medical treatment unless contraindicated and ticagrelor is preferred to clopidogrel. (COR I, LOE B)
- Prasugrel (60 mg loading dose, 10 mg daily dose) is recommended only in NSTEMI-ACS patients undergoing PCI without contraindication. (COR I, LOE B)
- Clopidogrel rather than ticagrelor or prasugrel may be considered in patients with increased bleeding risk features. (COR IIa, LOE C)
- Reduced dose of prasugrel (20 mg loading dose, 3.75 mg daily dose) may be considered in NSTEMI-ACS patients undergoing PCI if there are increased bleeding risk features. (COR IIb, LOE B)
- Pretreatment with prasugrel before diagnostic angiography is not recommended for NSTEMI-ACS patients. (COR III, LOE B)

Glycoprotein IIb/IIIa receptor inhibitors

Glycoprotein (GP) IIb/IIIa inhibitors lead to high degree of platelet inhibition and they were associated with reduced ischemic events in NSTEMI-ACS patients undergoing PCI.^{111–113} One meta-analysis involving six randomized control trials with 29,570 NSTEMI-ACS patients demonstrated that GP IIb/IIIa inhibitors significantly reduced the risk of 30-day death or MI and the benefit was more pronounced in patients undergoing PCI during hospitalization.¹¹⁴ In the Acute Catheterization and Urgent Intervention Triage Strategy (ACUITY) trial, the ischemic events (death, MI, unplanned revascularization) were similar between NSTEMI-ACS patients receiving bivalirudin alone and who received heparin plus GP IIb/IIIa inhibitors and there was fewer major bleeding risk in the bivalirudin group.¹¹⁵ The ACUITY Timing trial and the Early Glycoprotein IIb/IIIa Inhibition in Non-ST-Segment Elevation Acute Coronary Syndrome (EARLY ACS) trial evaluated the optimal timing of GP IIb/IIIa inhibitors initiation in NSTEMI-ACS patients. Both studies demonstrated that early routine administration of GP IIb/IIIa inhibitors before angiography only increased bleeding risk but without improving ischemic outcomes.^{116,117} The efficacy and safety of GP IIb/IIIa inhibitors in patients with prasugrel or ticagrelor have not been prospectively studied. At current stage, due to routine use of DAPT with potent P2Y₁₂ inhibitors, GP IIb/IIIa inhibitors is now only applied within the catheterization laboratory for management of coronary thrombotic lesions or for bailout usage in thrombotic complications after PCI.

Recommendation

- Use of GP IIb/IIIa inhibitors in patients with NSTEMI-ACS as adjunctive therapy during PCI may be indicated for large thrombus burden or as bailout for thrombotic complications. (COR IIa, LOE B)
- Routine use of GP IIb/IIIa inhibitors in patients with NSTEMI-ACS is not recommended before coronary angiography. (COR III, LOE A)

Anticoagulation therapy

Unfractionated heparin

Previous studies using unfractionated heparin (UFH) and aspirin in patients with NSTEMI-ACS were performed primarily in the era before DAPT and early revascularization. These studies found a tendency to reduce adverse cardiac events with the addition of UFH to aspirin.^{76,118,119} Currently, the recommended weight-adjusted dose of UFH is a loading dose of 60 IU/kg (maximum 4000 IU) with an infusion rate of 12 IU/kg/hr (maximum 1000 IU/hr) and adjusted by monitoring activated partial thromboplastin time (aPTT). It can be continued for 48 h or until PCI is completed.

Low-molecular-weight heparin

Low-molecular-weight heparin (LMWH) has a lower molecular weight about one third of UFH and has both anti-Xa and anti-IIa activity. LMWH is easily absorbed after subcutaneous (SC) injection. The anticoagulant activity of LMWH does not

require to be monitored routinely. The recommended dose of enoxaparin is 1 mg/kg SC injection every 12 h. In patients with creatinine clearance (CrCl) < 30 mL/min, the dose should be reduced to 1 mg/kg SC once daily or consider to use UFH. In the Efficacy and Safety of Subcutaneous Enoxaparin in Non-Q-Wave Coronary Events (ESSENCE) trial, the risk of composite endpoint (death, MI, or recurrent angina) and need for revascularization were significantly lower in NSTEMI-ACS patients treated with enoxaparin than in those with UFH.^{120,121} The risk of 30-day major bleeding was similar between enoxaparin and UFH. But the incidence of minor bleeding was significantly higher in the enoxaparin group due to ecchymosis at the injection sites. In the Superior Yield of the New Strategy of Enoxaparin, Revascularization and Glycoprotein IIb/IIIa Inhibitors (SYNERGY) trial of high risk NSTEMI-ACS patients treated with an early invasive strategy, the clinical outcomes of death or MI were similar at 30 days between enoxaparin versus UFH. There was more TIMI major bleeding in the enoxaparin group but no statistically significant increase in GUSTO (Global Utilization of Streptokinase and t-PA for Occluded Arteries) severe bleeding and transfusion.¹²² Enoxaparin should be continued for the duration of hospitalization or until PCI is completed.

Recommendation

- In patients with NSTEMI-ACS, anticoagulation is recommended for all patients with or without an initial invasive strategy if there is no contraindication. UFH or enoxaparin can be used. (COR I, LOE B)
- UFH is given at loading dose of 60 IU/kg (maximum 4000 IU) with initial infusion of 12 IU/kg/hr (maximum 1000 IU/hr) and adjusted by checking aPTT. (COR I, LOE A)
- Enoxaparin is given with 1 mg/kg SC every 12 h. A lower dose with 1 mg/kg SC once daily or use UFH in patients with CrCl < 30 mL/min (COR I, LOE B)

Fondaparinux

Fondaparinux is a selective inhibitor of activated factor X and is mainly excreted by the kidney. It is contraindicated in patients with CrCl < 30 mL/min. In patients with NSTEMI-ACS, the dose of fondaparinux is 2.5 mg SC daily for the duration of hospitalization up to 8 days or until PCI is performed. In the Fifth Organization to Assess Strategies in Acute Ischemic Syndromes (OASIS-5) trial, patients with NSTEMI-ACS were randomized to receive either fondaparinux (2.5 mg SC daily) or enoxaparin (1 mg/kg SC twice daily) for 8 days.¹²³ The primary endpoint was similar at 9 days between the two groups, but major bleeding was significantly less in the fondaparinux group. In order to avoid catheter-related thrombosis in NSTEMI-ACS patients undergoing PCI when fondaparinux is used alone, an anticoagulant with anti-IIa activity (UFH or bivalirudin) should be used during PCI.^{124,125}

Recommendation

- In NSTEMI-ACS, fondaparinux 2.5 mg SC daily can be considered to continue up to 8 days of hospitalization or until PCI is performed. (COR IIa, LOE B)
- If PCI is performed in patients treated with fondaparinux, additional anticoagulant with UFH or bivalirudin

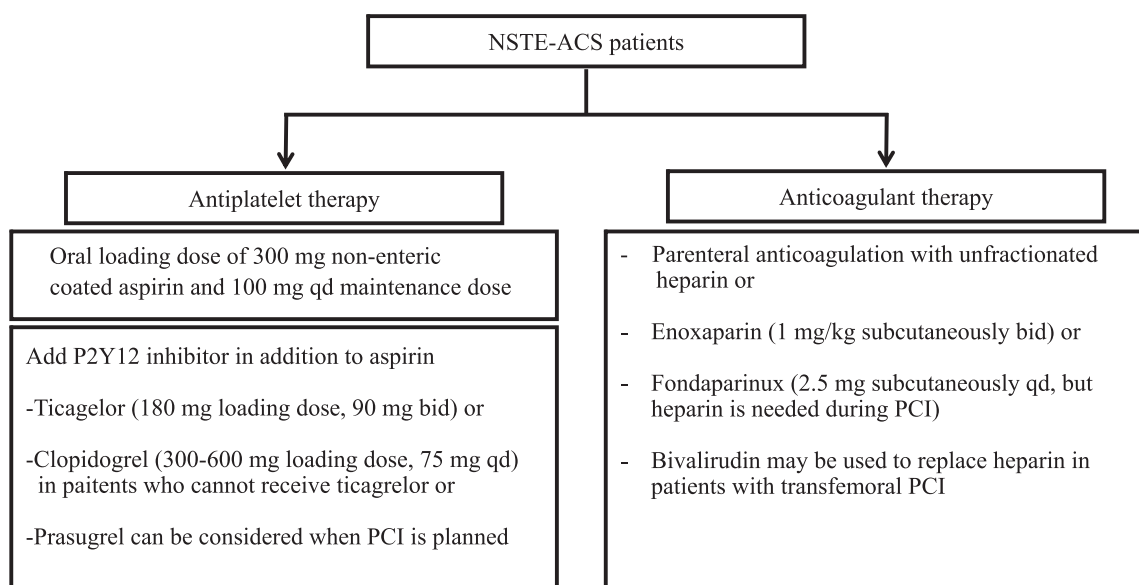


Figure 3 Recommendations of antiplatelet and anticoagulant treatment for NSTEMI-ACS. NSTEMI-ACS, non-ST-segment elevation acute coronary syndrome; PCI, percutaneous coronary intervention.

should be used to reduce the risk of catheter-related thrombosis. (COR IIa, LOE B)

- In NSTEMI-ACS, bivalirudin may be considered to replace UFH for patients requiring transfemoral PCI. (COR IIa, LOE B)

Bivalirudin

Bivalirudin was used in the ACUTY trial in 13,819 moderate- to high-risk patients with NSTEMI-ACS with an invasive strategy. The patients were randomized into three major groups including either UFH or LMWH with a GP IIb/IIIa inhibitor, bivalirudin with a GP IIb/IIIa inhibitor, or bivalirudin alone. Most patients received 300 mg clopidogrel loading before intervention. Bivalirudin alone was noninferior to UFH/LMWH combined with a GP IIb/IIIa inhibitor in the composite ischemia endpoint (7.8% vs. 7.3%; HR 1.08). There was a significantly lower rate of major bleeding (HR 0.53, $p < 0.001$) when bivalirudin was used alone.¹¹⁵ The combination of bivalirudin with a GP IIb/IIIa inhibitor showed similar ischemic outcomes compared with bivalirudin alone but had higher rate of major bleedings. A clinical study to compare bivalirudin with UFH monotherapy based on current common practice with radial access for PCI, potent P2Y12 inhibitors and without planned use of GP IIb/IIIa inhibitors was published recently. In 6006 patients with acute MI (STEMI or NSTEMI), there was no significant difference in death or MI between bivalirudin and UFH. The major bleeding events were also similar between treatment groups.¹²⁶ A recent observation study of NSTEMI also demonstrated that the reduction of bleeding events of bivalirudin was observed only in patients undergoing transfemoral PCI and no bleeding benefit was observed for transradial PCI.¹²⁷ Fig. 3 summarizes the current recommendations of antiplatelet and anticoagulant treatment during admission for NSTEMI-ACS in Taiwan.

Recommendation

Revascularization strategy

Timing of invasive coronary angiography

Patients with NSTEMI-ACS should triage first according to their initial clinical presentation into different risk groups by using TIMI score or GRACE score. For low risk patients, conservative medical treatment could be considered first. However, for those patients with high risk features and risk scores, early invasive coronary angiography becomes important. The studies on the optimal timing of coronary angiography in NSTEMI-ACS patients showed conflicting results mainly due to different risk profiles of the studied populations. A meta-analysis of NSTEMI-ACS that included seven randomized clinical trials with 5370 patients and four observational studies with 77,499 patients compared an early (<24 h) with a delayed invasive strategy showed no significant benefit for death (odds ratio [OR] 0.83, 95% CI 0.64–1.09), MI (OR 1.15, 95% CI 0.65–2.01) or major bleeding (OR 0.76, 95% CI 0.56–1.04).¹²⁸ Only a lower risk of refractory ischemia (OR 0.55, 95% CI 0.35–0.86) can be seen in early invasive strategy. The Timing of Intervention in Acute Coronary Syndromes (TIMACS) study is the largest randomized trial and assigned 3031 NSTEMI-ACS patients to an early (<24 h, median time 14 h) or delayed (>36 h, median time 50 h) coronary angiography. At 6 months, the primary composite endpoint of death, MI or stroke was not different between the early and delayed invasive strategy (HR 0.85, 95% CI 0.68–1.06). The secondary endpoint of death, MI, stroke or refractory ischemia was reduced by 28% in favor

of the early invasive strategy (HR 0.72, 95% CI 0.58–0.89). In high-risk patients with GRACE risk score >140 , an early invasive strategy lowered the risk of death, MI or stroke (HR 0.65, 95% CI 0.48–0.89).¹²⁹ In a post hoc analysis of the ACUITY trial, a delay to PCI > 24 h was an independent predictor of 30-day and 1-year mortality. The excess of ischemic events associated with the PCI >24 h strategy was most apparent among moderate- and high-risk patients.¹³⁰ In the NSTEMI-ACS patients undergoing PCI from the Taiwan ACS Full Spectrum Registry, early PCI within 24 h could not improve the primary outcome (cardiac death and recurrent MI) compared with late PCI (24–72 h or > 72 h) in the low (TIMI risk score 0–2) and intermediate (TIMI risk score 3–4) risk groups. In the high risk group (TIMI risk score 5–7), patients who underwent PCI after 72 h had significantly worse primary outcomes than those who underwent PCI within 24–72 h.¹³¹

It is generally agreed that the higher the risk and the more unstable the condition, the sooner the invasive strategy should be adopted. It is not recommended to perform coronary angiography routinely in all NSTEMI-ACS patients. For low risk patients without symptoms after initial medical treatment, coronary angiography can be delayed and reserved for patients with positive stress test or recurrent symptoms. However, in the very high risk patients, such as unstable hemodynamics, acute pulmonary edema, severe tachy- or bradyarrhythmias, and refractory angina with dynamic ECG changes, early coronary angiography should be considered and performed as soon as possible within 24 h. For all the other patients, coronary angiography can be considered within 24–72 h after intensified anti-thrombotic and optimal anti-ischemic medical treatment.

Recommendation

- **An early coronary angiography should be considered and performed as soon as possible within 24 h for NSTEMI-ACS patients who have unstable hemodynamics, acute pulmonary edema, tachy- or bradyarrhythmias and refractory angina with dynamic ECG changes. (COR I, LOE B)**

Revascularization therapy

Coronary angiography in NSTEMI-ACS patients usually showed multivessel disease. If the culprit lesions can be identified and coronary anatomy is suitable, ad hoc PCI should be performed especially in the very high risk patients with unstable hemodynamics, acute pulmonary edema, tachy- or bradyarrhythmias or refractory angina with occlusion of the culprit artery. In addition to the culprit lesions, to treat all the other significant lesions with PCI or CABG for complete revascularization in one setting or to stage the procedures should depend on the clinical conditions and coronary angiographic complexities of the patient. The Heart Team in the institution should decide the best revascularization strategy for each NSTEMI-ACS patient with multivessel disease and the adequate timing of CABG if necessary. Several risk scores have been developed to stratify patients at high risk of revascularization for left main or multivessel disease. The Synergy between PCI with

Taxus and Cardiac Surgery (SYNTAX) score is commonly used to represent angiographic complexity which is considered an important determinant of outcomes after PCI or CABG.^{132–134} In patients with high SYNTAX score (≥ 33), the clinical outcomes of PCI were shown to be inferior to CABG and there was no significant difference between PCI and CABG among patients with SYNTAX score < 33 .¹³² In addition to the extent and complexity of CAD, completeness of revascularization, long-term durability of vessel patency, procedure risk of PCI, and operative mortality should be taken into consideration. The clinical characteristics of the patient including the presence of DM, CKD, cardiac function, previous CABG history and the ability of the patient to tolerate DAPT should also be considered. By using shared decision making, there should be a fully discussion about the benefit and risk of CABG or multivessel PCI between patients, their families and physicians.

Recommendation

- **For NSTEMI-ACS patients with multivessel disease and the culprit lesions can be identified clearly, ad hoc PCI could be performed especially in very high risk patients with unstable hemodynamics, acute pulmonary edema, tachy- or bradyarrhythmias or refractory angina with occlusion of the culprit artery. (COR I, LOE C)**
- **For NSTEMI-ACS patients with multivessel disease, the Heart Team should use shared decision making to decide the revascularization strategy. (COR I, LOE C)**

PCI strategy

Radial artery has become the vascular access of choice during PCI. In the Minimizing Adverse Haemorrhagic Events by Transradial Access Site and Systemic Implementation of angioX (MATRIX Access) study, a total of 8404 ACS patients undergoing PCI were randomly assigned to radial or femoral access. Two prespecified coprimary 30-day composite outcomes were: major adverse CV events, defined as the composite of all-cause mortality, MI, or stroke; and net adverse clinical events, defined as the composite of major bleeding not related to CABG or major adverse CV events. Patients with radial access had lower risk of major adverse CV events compared with femoral access (risk ratio [RR] 0.85, 95% CI 0.74–0.99). The net adverse clinical events were also reduced in radial access group (RR 0.83, 95% CI 0.73–0.96). The difference was mainly driven by reduction of major bleeding and all-cause mortality in the radial access patients.¹³⁵ The Radial Vs femoral access for coronary intervention (RIVAL) trial randomly allocated 7021 ACS patients to radial or femoral artery access. The primary outcome was a composite of death, MI, stroke or non-CABG-related major bleeding at 30 days, which occurred in 128 (3.7%) of 3507 patients in the radial access group compared with 139 (4.0%) of 3514 in the femoral access group (HR: 0.92, 95% CI: 0.72–1.17).¹³⁶ But the overall PCI volume and radial PCI volume of the center were independently associated with risk reduction of the primary outcome in a multivariable model.¹³⁷ Therefore, the radial access is recommended to be performed by an experienced operator

at high volume center. For selection of stents, based on previous PCI studies that enrolled ACS patients, new generation drug-eluting stent (DES) is recommended over bare metal stent (BMS) because of reduced rates of restenosis and target lesion revascularization.^{138–140}

Recommendation

- In NSTEMI-ACS patients, radial approach is recommended for coronary angiography and PCI in experienced operators and high volume centers with radial access (COR I, LOE A).
- New generation DESs are recommended in NSTEMI-ACS patients undergoing PCI. (COR I, LOE A).

Timing of CABG

PCI is usually the initial procedure of choice for immediate revascularization in patients with NSTEMI-ACS. Emergent CABG in NSTEMI-ACS is usually reserved for patients with PCI-associated complications or rare MI-related mechanical complications. For stable NSTEMI-ACS patients that need CABG, the optimal timing should be determined individually because of the absence of randomized clinical trial data. Critical patients with severe coronary stenosis or frequent recurrent angina should receive CABG as soon as possible without considering platelet function following discontinuation of DAPT. For all the other stable patients, CABG appears to be most beneficial when surgery is performed after several days of medical stabilization and discontinuation of DAPT. The optimal timing of CABG after acute MI is still controversial. In the retrospective review of California discharge data, the mortality was higher in acute MI patients who received early (<3 days) than delayed (≥3 days) CABG (propensity-adjusted OR 1.40; 95% CI 1.12–1.74).¹⁴¹ In the Alberta Provincial Project for Outcomes Assessment in Coronary Heart Disease (APPROACH) database, there was a nonsignificant trend to increased mortality for those undergoing early (2–7 days) (HR 2.36, 95% CI 0.72–7.76) and intermediate (8–14 days) (HR 1.68, 95% CI 0.76–3.72) than late CABG surgery (15–60 days).¹⁴² In the Acute Coronary Treatment and Intervention Outcomes Network Registry—Get with the Guidelines (ACTION Registry-GWTG) study, there were no significant differences of in-hospital mortality and the composite endpoint of death, MI, congestive heart failure or cardiogenic shock between the patients received early (≤48 h of admission) and late surgery (>48 h) after admission for NSTEMI.¹⁴³ Bleeding is the major concern when patients are under antiplatelet therapy. However, aspirin usually could be continued throughout surgery. In patients with increased bleeding risk, aspirin is advised to stop for 3–5 days before surgery. For P2Y₁₂ inhibitor, clopidogrel should be stopped for 5 days and prasugrel for 7 days before surgery. Recent study demonstrated that platelet function could recover to an acceptable level for surgery after discontinuation of ticagrelor for 72 h in the majority of patients.¹⁴⁴ The risk of CABG-related bleeding was also similar in ticagrelor-treated patients when the drug was discontinued between 72 and 120 h or >120 h before

surgery.¹⁴⁵ So surgery can be performed after discontinuation of ticagrelor for at least 3 days.

Recommendation

- In unstable NSTEMI-ACS patients with an indication of emergent CABG, it should be performed regardless of antiplatelet therapy. (COR IIa, LOE C)
- For stable NSTEMI-ACS patients that need CABG, CABG could be performed after several days of medical stabilization and discontinuation of DAPT. (COR I, LOE B)
- In patients for elective CABG, ticagrelor should be discontinued for at least 3 days. Clopidogrel should be discontinued for at least 5 days. Prasugrel should be discontinued for at least 7 days before surgery. (COR I, LOE B)

Long term management

Medical therapies for secondary prevention

The goals of therapy after NSTEMI-ACS are to prevent recurrent ischemic events and improve survival. Evidence-based care should be given and risk factors should be treated aggressively through lifestyle modification and pharmacological therapies according to current guidelines in Taiwan. In most situations, DAPT, beta blockers, statins, and inhibitors of the renin-angiotensin-aldosterone system given during hospitalization should be continued after discharge because of the prognostic benefits. Each patient should be treated individually on the basis of underlying diseases and in-hospital interventions such as PCI or CABG. The mnemonic the “ABCDE” (“A” for antiplatelet therapy, ACEI/ARB, and aldosterone antagonist; “B” for beta blockers and blood pressure control; “C” for cholesterol treatment and cigarette smoking cessation; “D” for diabetes management and diet; and “E” for exercise) is beneficial in guiding comprehensive management.¹⁴⁶

Recommendation

- ACEIs (or ARBs), beta blockers, and statins (or statin/ezetimibe) should be continued in NSTEMI-ACS patients after discharge if there is no contraindication. (COR I, LOE A).
- Aspirin should be continued indefinitely and the maintenance dose should be 100 mg daily in NSTEMI-ACS patients. (COR I, LOE A)

Duration of DAPT

In addition to aspirin, P2Y₁₂ inhibitors should be continued for at least 12 months after discharge in NSTEMI-ACS patients received PCI, CABG or only medical treatment. The trade-off between ischemic and bleeding risk should be considered when using longer duration of DAPT more than 12 months.^{147–149} In the Taiwan ACS Full Spectrum Registry, bleeding events contributed to higher rate of major CV outcome at 1-year follow-up.¹⁵⁰ Therefore, decisions about duration of DAPT should be individually made and physicians should integrate clinical findings and assessment of the ischemic/bleeding risk.

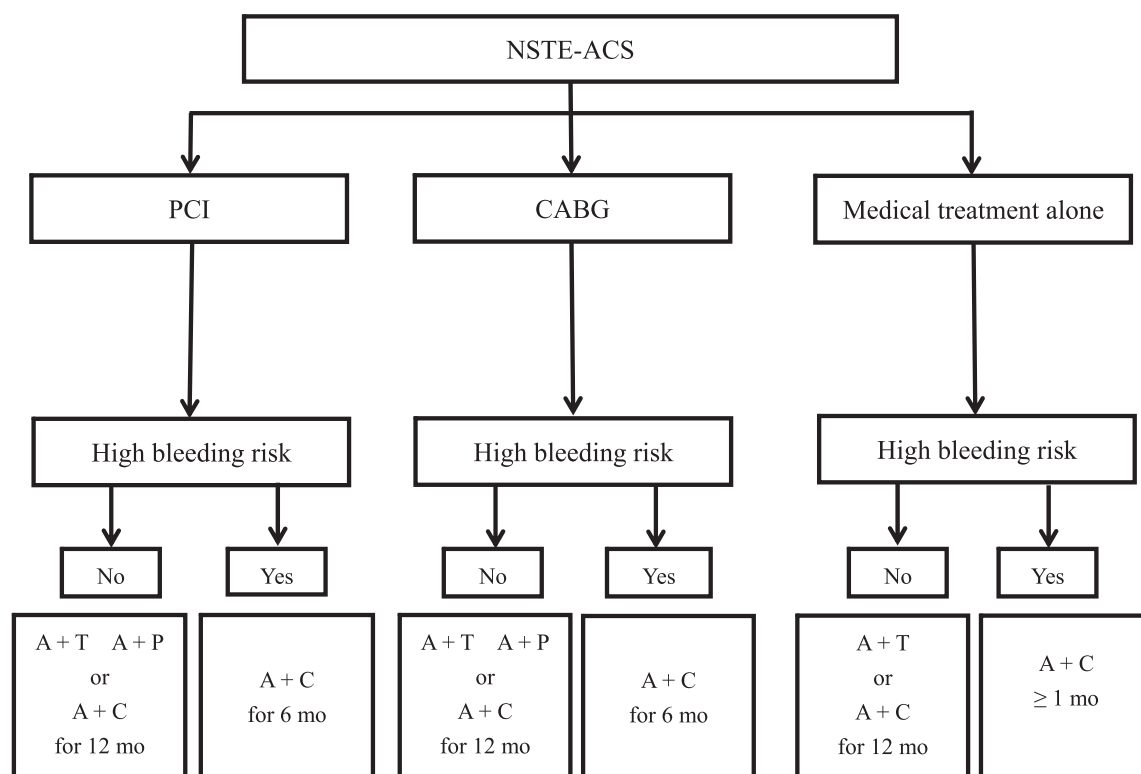


Figure 4 Recommendations of dual antiplatelet therapy duration for NSTEMI-ACS patients. A + C, aspirin plus clopidogrel; A + P, aspirin plus prasugrel; A + T, aspirin plus ticagrelor; mo, month; NSTEMI-ACS, non-ST-segment elevation acute coronary syndrome. A + T or A + P is preferred to A + C in patients without high bleeding risk features.

DAPT duration in patients treated with PCI

In NSTEMI-ACS patients receiving PCI, DAPT with aspirin and clopidogrel has been recommended for 1 year over aspirin alone according to the PCI-CURE study,⁸⁴ while the TRITON-TIMI 38 and PLATO studies have demonstrated the superiority of a prasugrel- and ticagrelor-based regimen, respectively, over a clopidogrel-based one.^{95,100} The post hoc analysis of the Clopidogrel for High Atherothrombotic Risk and Ischemic Stabilization, Management, and Avoidance (CHARISMA) study showed that DAPT with aspirin and clopidogrel decreased the composite endpoint of CV death/MI/stroke in patients with prior MI, ischemic stroke, or symptomatic PAD.¹⁴⁷ High risk patients with 1–3 years after MI in the Prevention of Cardiovascular Events in Patients with Prior Heart Attack Using Ticagrelor Compared to Placebo on a Background of Aspirin–Thrombolysis in Myocardial Infarction 54 (PEGASUS-TIMI 54) trial were randomized to either DAPT (with ticagrelor 60 mg or 90 mg twice daily) or continued aspirin monotherapy.¹⁴⁹ After a mean of 33-month follow up, DAPT resulted in a 1.2–1.3% absolute reduction in the CV death/MI/stroke but a 1.2–1.5% absolute increase in major bleeding, with no excess in fatal bleeding or ICH. About 83% patients had prior PCI history in the PEGASUS-TIMI 54 study. In the Dual Antiplatelet Therapy (DAPT) study, the ischemic benefit/bleeding risk ratio for prolonged DAPT was more favorable for those presenting with MI than those with stable CAD.¹⁴⁸ In a meta-analysis of patients with prior MI enrolled in 6 randomized clinical trials, prolonged DAPT significantly decreased the

absolute risk of major adverse CV events by 1.1% and significantly increased the absolute risk of major bleeding by 0.8%.¹⁵¹ In general, prolonged DAPT may be more favorable for those with prior MI, with an absolute decrease in ischemic events of about 1%–3% and an absolute increase in bleeding events of about 1% (median duration of DAPT therapy: 18–33 months). The Taiwan ACS Full Spectrum Registry (3131 ACS patients, 46.8% NSTEMI-ACS) reported that the DAPT (aspirin and clopidogrel) prescribed to 74.8% patients at discharge and declined to only 24.9% patients a year later. The probability of death/MI/stroke was higher in patients who received DAPT for less than 9 months (OR 1.61, $p = 0.011$).⁸

Recommendation

- In NSTEMI-ACS patients receiving PCI, P2Y₁₂ inhibitor therapy should be given for at least 12 months if there is no bleeding complication. (COR I, LOE A)
- In patients with NSTEMI and high ischemic risk, DAPT can be used longer than 12 months. (COR IIa, LOE B)
- In prolonged DAPT treatment longer than 12 months, ticagrelor 60 mg twice daily may be considered. (COR IIb, LOE B)

DAPT duration in patients treated with CABG

In the subgroup analysis of the CURE study, patients who underwent CABG treated with DAPT had lower ischemic events compared with aspirin monotherapy users, which

was consistent with the overall population.⁷⁹ In the largest meta-analysis of patients pooled from 5 randomized clinical trials and 6 observational studies, DAPT (aspirin and clopidogrel) was associated with lower vein graft occlusion and 30-day mortality rate as compared with aspirin monotherapy.¹⁵² However, major bleeding after surgery was more frequent with DAPT. The TRITON-TIMI 38 subgroup analysis showed that prasugrel treatment was associated with a significantly lower 30-day mortality than that of clopidogrel and more postoperative blood loss in CABG patients.¹⁵³ Similarly, patients who underwent CABG in the PLATO study had similar primary endpoint at 1 year for both groups, but a significant reduction in CV mortality was noted with ticagrelor compared with clopidogrel.¹⁰⁰

Recommendation

- In NSTEMI-ACS patients who underwent CABG, P2Y₁₂ inhibitors should be resumed after CABG to complete 12-month DAPT therapy. (COR I, LOE B)

DAPT duration in patients treated with medical therapy alone without revascularization

DAPT comprising aspirin and clopidogrel in CURE study has reduced recurrent ischemic events in the NSTEMI-ACS patients compared with aspirin alone irrespective of revascularization strategy.⁷⁹ The TRILOGY ACS study randomized 7243 NSTEMI-ACS patients receiving only medical management to clopidogrel or prasugrel for a median of 17 months.⁹⁶ Prasugrel did not reduce the CV death/MI/stroke (HR 0.91, 95% CI 0.79–1.05) compared with clopidogrel. In patients <75 years old, the TIMI major and minor bleeding events were more frequent in the prasugrel group (HR 1.54, CI 1.06–2.23). In the NSTEMI-ACS patients managed with or without revascularization in the PLATO trial, the CV death/MI/stroke was significantly reduced with ticagrelor compared with clopidogrel (HR 0.83, CI 0.74–0.93).¹⁰³ Major bleeding rate was similar between groups (HR 1.07, 95% CI 0.95–1.19), but ticagrelor was associated with an increase in non-CABG major bleeding (HR 1.28, 95% CI 1.05–1.56). Fig. 4 summarizes the recommendations of DAPT duration in NSTEMI-ACS patients.

Recommendation

- It is preferred to use ticagrelor instead of clopidogrel in NSTEMI-ACS patients with medical therapy for 12 months. (COR IIa, LOE B)
- Prasugrel is not suggested in NSTEMI-ACS patients with medical therapy. (COR III, LOE B)

Factors associated with increased ischemic and bleeding risk

Factors that have been associated with increased ischemic risk (including stent thrombosis) and increased bleeding risk were proposed in many studies.^{154–157} Generally speaking, old age, CKD, ACS presentation, repeated ACS, extensive CAD, and DM were regarded as risk factors of recurrent ischemic events. Other risk factors are associated with increased stent thrombosis including DM, heart failure with reduced left ventricular ejection fraction, and complex PCI

procedures, such as 2-stent deployment at the bifurcation lesions, small stent size, and greater stent length. Some factors are associated with both increased ischemic and bleeding risk, making it difficult to decide the duration of DAPT. A new risk score (the DAPT score), derived from the DAPT study, can be considered in deciding the DAPT duration.¹⁵⁸ The data suggest that in patients treated for 1-year DAPT without significant bleeding or ischemic events, prolonged DAPT (>1 year) may be favorable for those with a high DAPT score (≥2). On the contrary, standard DAPT duration should be used in those with a low DAPT score (<2) (Table 7).

Recommendation

- If the bleeding risk outweighs the benefit of 12-month DAPT, earlier discontinuation (< 12 months) of P2Y₁₂ inhibitor therapy can be considered. (COR IIb, LOE C)
- The DAPT score can be used to evaluate the benefit of prolonged DAPT therapy. (COR IIb, LOE B)

NSTEMI-ACS patients that need long term oral anticoagulation

Treating patients with NSTEMI-ACS that need long term OAC therapy is challenging. AF occurs in 6–21% in patients hospitalized with acute MI.¹⁵⁹ AF not only increases the stroke risk in hospitalized patients with acute MI but also during follow-up after discharge.^{160,161} AF is a common indication for OAC in NSTEMI-ACS patients. Cardiac thrombosis, mechanical heart valves and venous thromboembolism are other reasons that OAC becomes necessary in NSTEMI-ACS patients. DAPT with aspirin and P2Y₁₂ inhibitor is more effective than OAC to improve ischemic outcomes in patients with ACS and/or received coronary stents; while OAC is better than DAPT in prevention of AF-related

Table 7 The DAPT score.

DAPT score	
After 12 months of uneventful DAPT	
Standard DAPT (12 months) vs. Long DAPT (30 months)	
Age (years)	
≥ 75	–2 points
65 to < 75	–1 point
< 65	0 point
Current cigarette smoking	+1 point
Diabetes mellitus	+1 point
MI at presentation	+1 point
Prior PCI or prior MI	+1 point
Paclitaxel-eluting stent	+1 point
Stent diameter < 3 mm	+1 point
HF or LVEF < 30%	+2 points
Vein graft PCI	+2 points
–2 to 10 points	
Score ≥ 2 → Long DAPT	
Score < 2 → Standard DAPT	

DAPT, dual antiplatelet therapy; HF, heart failure; LVEF, left ventricular ejection fraction; MI, myocardial infarction.

PCI, percutaneous coronary intervention.

Adapted from reference.¹⁵⁸

stroke.^{162,163} Combining DAPT with OAC, including vitamin K antagonist (VKA) or non-vitamin K antagonist oral anticoagulant (NOAC), increases bleeding risk. In patients with AF, triple therapy with warfarin, aspirin, and clopidogrel had higher bleeding risk compared with warfarin monotherapy or DAPT.¹⁶⁴ In an observational study of 12,165 patients with AF hospitalized with MI and/or PCI from the Danish National Patient Registry, clopidogrel plus VKA was found to be equal to or better than VKA plus DAPT on both efficacy and safety outcomes.¹⁶⁵ The What is the Optimal antiplatelet and anticoagulant therapy in patients with oral anticoagulation and coronary Stenting (WOEST) trial was the first randomized, controlled study to compare clopidogrel plus VKA with triple therapy of DAPT plus VKA.¹⁶⁶ It included patients who took VKA for various reasons and underwent PCI. Among them, 69% of patients had AF. This trial found that clopidogrel plus VKA reduced the risk of bleeding compared with triple therapy of DAPT plus VKA. The thrombotic risks in clopidogrel plus VKA group, including stroke, MI and stent thrombosis were similar compared with DAPT plus VKA group.¹⁶⁶ The Open-Label, Randomized, Controlled, Multicenter Study Exploring Two Treatment Strategies of Rivaroxaban and a Dose-Adjusted Oral Vitamin K Antagonist Treatment Strategy in Subjects with Atrial Fibrillation who Undergo Percutaneous Coronary Intervention (PIONEER AF-PCI) study was the first randomized clinical trial to compare NOAC (rivaroxaban 15 mg qd) plus P2Y12 inhibitor with the traditional triple therapy of VKA plus DAPT in AF patients that need PCI with stent placement.¹⁶⁷ Among the study participants, 52% were ACS and 66% used DES. Overall, bleeding risk was significantly reduced with rivaroxaban plus clopidogrel compared with VKA plus DAPT group. The efficacy endpoints including major adverse CV event or stent thrombosis were similar between the groups. The Randomized Evaluation of Dual Antithrombotic Therapy with Dabigatran versus Triple Therapy with Warfarin in Patients with Nonvalvular Atrial Fibrillation Undergoing Percutaneous Coronary Intervention (RE-DUAL PCI) study also evaluated the NOAC (dabigatran 110 and 150 mg bid) plus P2Y12 inhibitor with the traditional triple therapy of VKA plus DAPT in AF patients underwent PCI.¹⁶⁸ In the RE-DUAL PCI study, 50.5% patients were ACS and more than 80% received DES. In P2Y12 inhibitors, most patients used clopidogrel and 12% received ticagrelor. The risk of bleeding was lower in dabigatran plus P2Y12 inhibitor than triple therapy with VKA and DAPT. The efficacy endpoint (thromboembolic events, death or unplanned revascularization) was noninferior to triple therapy.¹⁶⁸ Although the case numbers in these clinical trials were underpowered to definitively confirm the efficacy outcomes, the PIONEER AF-PCI and RE-DUAL PCI studies carried important clinical implication that P2Y12 inhibitor plus an NOAC maybe enough in AF patients that need PCI with stent placement. Other clinical trials of NOAC using apixaban or edoxaban are still undergoing to compare the safety and efficacy of various NOAC plus P2Y12 inhibitor with VKA plus DAPT in patients with AF and need PCI. The results of these studies may give the definite answer whether NOAC plus P2Y12 inhibitor is the best choice for these patients.

The CHA2DS2-VASc score (Table 8) ≥ 2 usually indicates significantly increased embolic risk of AF.¹⁶⁹ The optimal

Table 8 The CHA2DS2-VASc score.

Risk factor	Score
^a Congestive heart failure/Left ventricular dysfunction	1
Hypertension	1
Age ≥ 75 years	2
Diabetes mellitus	1
Stroke/Transient ischemic attack/Thromboembolism	2
Vascular disease (prior myocardial infarction, peripheral artery disease, or aortic plaque)	1
Age 65–74 years	1
Sex category (i.e. female gender)	1

^a Documented moderate-to-severe systolic dysfunction [i.e. heart failure with reduced ejection fraction $<40\%$ (HF-REF)] or patients with recent decompensated heart failure requiring hospitalization, irrespective of ejection fraction [i.e. both HF-REF and heart failure with preserved ejection fraction (HF-PEF)].

Adapted from reference.¹⁶⁹

antithrombotic regimen in patients with NSTEMI-ACS and AF should depend on physicians' clinical judgment of the patient's risk (Fig. 5). For patients with NSTEMI-ACS and AF with CHA2DS2-VASc score ≥ 2 and ischemic risk $>$ bleeding risk, triple therapy with DAPT and NOAC can be used for 1–6 months. Then, dual therapy with NOAC and clopidogrel is used up to 12 months. When the bleeding risk is higher than ischemic risk, use NOAC plus clopidogrel without aspirin to replace triple therapy for 12 months. One year after NSTEMI-ACS, the patients are treated as stable CAD and NOAC monotherapy alone is used for long term antithrombotic treatment. Prolonged NOAC plus clopidogrel >12 months could be considered in patients with very high ischemic risk or received complex PCI procedures. For patients with NSTEMI-ACS and AF with CHA2DS2-VASc score <2 , DAPT is suggested for 12 months and use aspirin for long term treatment after 1 year. When combining NOACs with DAPT or clopidogrel, the lowest NOAC dose that has indication of stroke prevention for AF in Taiwan should be used. For patients with moderate or severe mitral stenosis, mechanical cardiac valve or severe CKD (CrCl < 15 mL/min for rivaroxaban or < 30 mL/min for dabigatran) that NOACs cannot be used, VKA with a lower PT INR level (around 2.0) should be considered when combining with antiplatelet agents.

Recommendation

- For NSTEMI-ACS patients with AF and CHA2DS2-VASc score ≥ 2 , triple therapy with DAPT and NOAC could be used for 1–6 months. (COR IIa, LOE C)
- For NSTEMI-ACS patients with AF at high bleeding risk, use dual therapy with NOAC plus clopidogrel to replace triple therapy for 12 months. (COR IIa, LOE B)
- For NSTEMI-ACS patients with AF, antiplatelet agents could be stopped 1 year after NSTEMI-ACS and NOAC

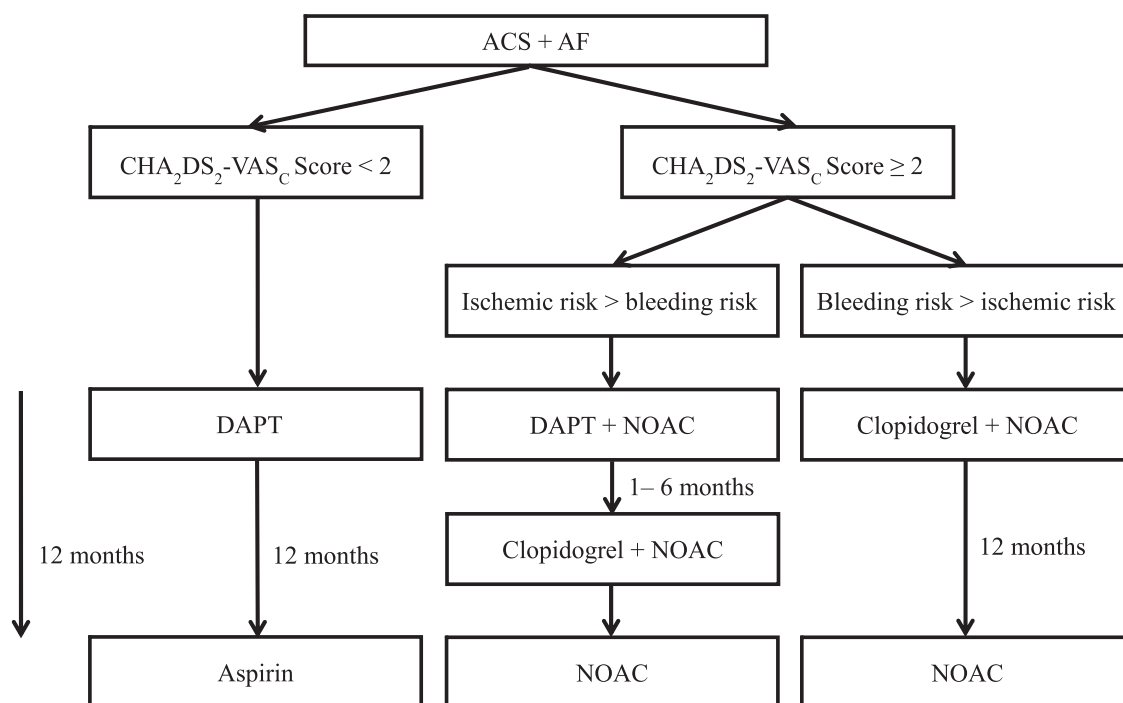


Figure 5 Recommendations of the optimal antithrombotic regimen and duration in patients with NSTEMI-ACS and AF. VKA should be used to replace NOAC in patients with moderate or severe mitral stenosis, mechanical cardiac valve or severe CKD (CrCl < 15 mL/min for rivaroxaban or <30 mL/min for dabigatran). AF, atrial fibrillation; CKD, chronic kidney disease; DAPT, dual antiplatelet therapy; NOAC, non-vitamin K antagonist oral anticoagulant; NSTEMI-ACS, non-ST-segment elevation acute coronary syndrome; VKA, vitamin K antagonist.

monotherapy should be continued indefinitely. (COR IIa, LOE C)

- For NSTEMI-ACS patients with AF at high ischemic risk or received complex PCI procedures, dual therapy with NOAC plus clopidogrel could be used longer than 12 months. (COR IIa, LOE C)

Conflicts of interest

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